Lab 07 – Cyber-HAZOP Workshop Simulation

Industrial Security (Master) • Maps to Section 07 • 4 hours, group of 4–6

LAB 07

Goal

Run a 4-hour facilitated Cyber-HAZOP on a P&ID provided by the instructor. Produce a documented findings file that a real plant could action.

Step Roles

- Facilitator (timekeeper, scribe)
- Process safety engineer
- Control / automation engineer
- Cyber-security architect
- Operations representative

Step 1 – Node decomposition

Take the provided P&ID. Break the system into nodes (a reactor vessel, a piping run, a heat exchanger, ...). One node = one workshop pass.

Step 2 – Apply guide words

For each node, walk through the HAZOP guide words (No, More, Less, Reverse, As well as, Part of, Other than). For each meaningful deviation, brainstorm:

- One classical safety cause (mechanical failure, operator error).
- One cyber cause (HMI compromise, PLC reprogram, network-borne).

Step 3 – Existing safeguards + gaps

For each cause: which safeguards already exist? Which gaps remain? Avoid the trap of "add MFA" – specify the vendor mechanism.

Step 4 – Findings file

Produce a worksheet (CSV / spreadsheet) with one row per cause, columns: node, guide-word, deviation, cause, safety/cyber, safeguard, gap, owner, due-date.

Deliverable

The completed worksheet and a 1-page chairman's summary identifying the top three actionable findings.

Caution

Fictional plant only. If your team accidentally produces a register that matches a real client's plant, redact and discard.